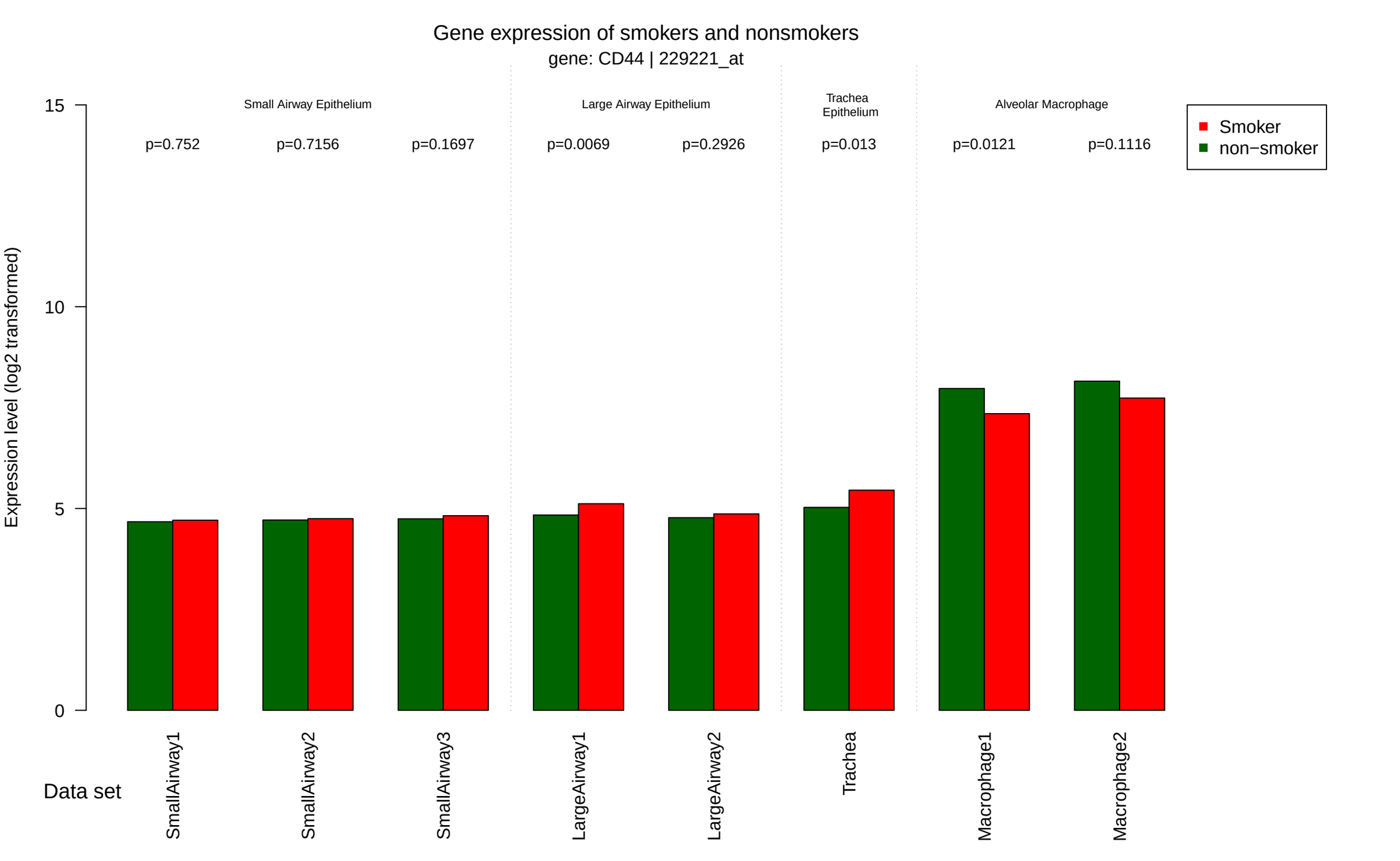
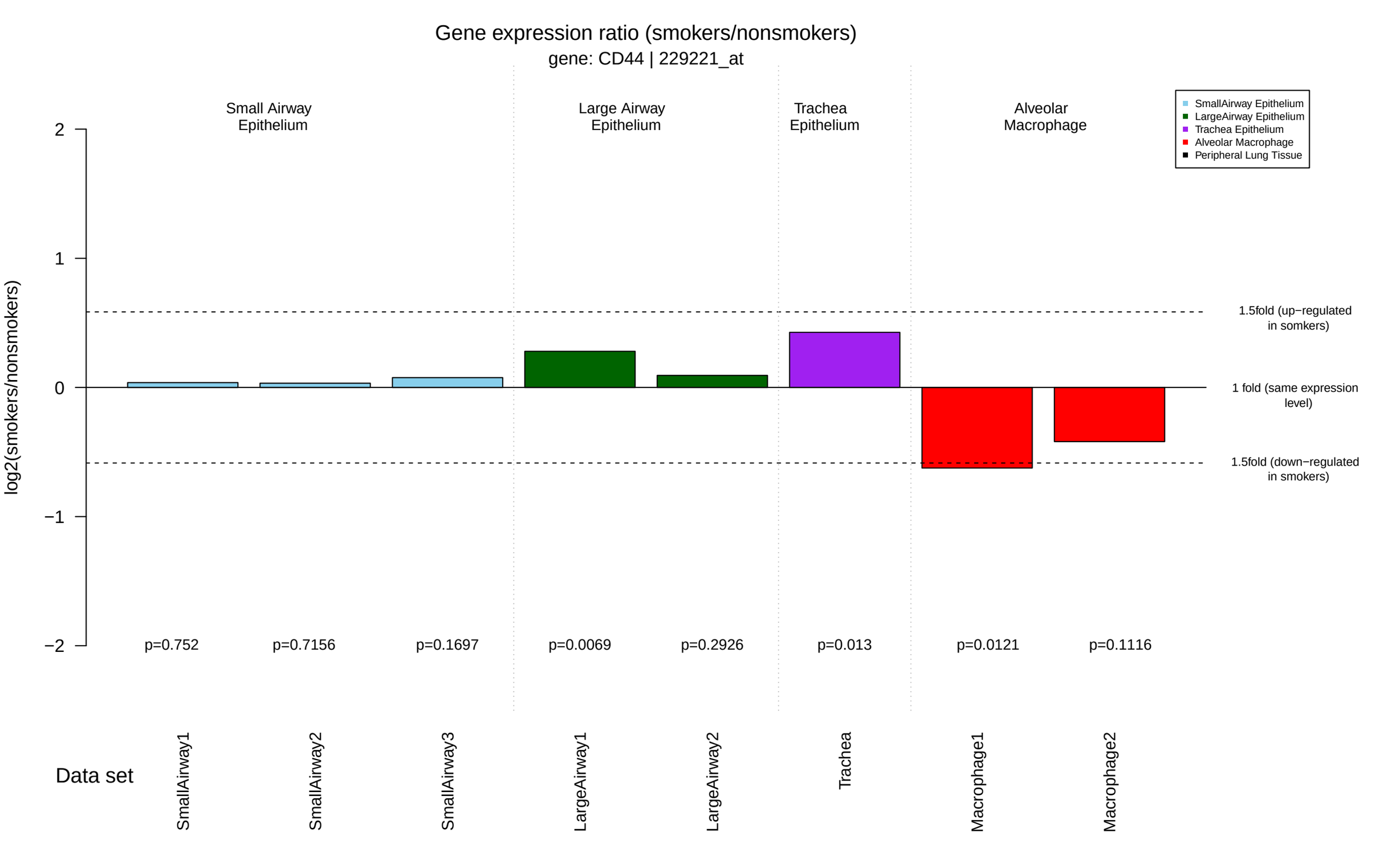
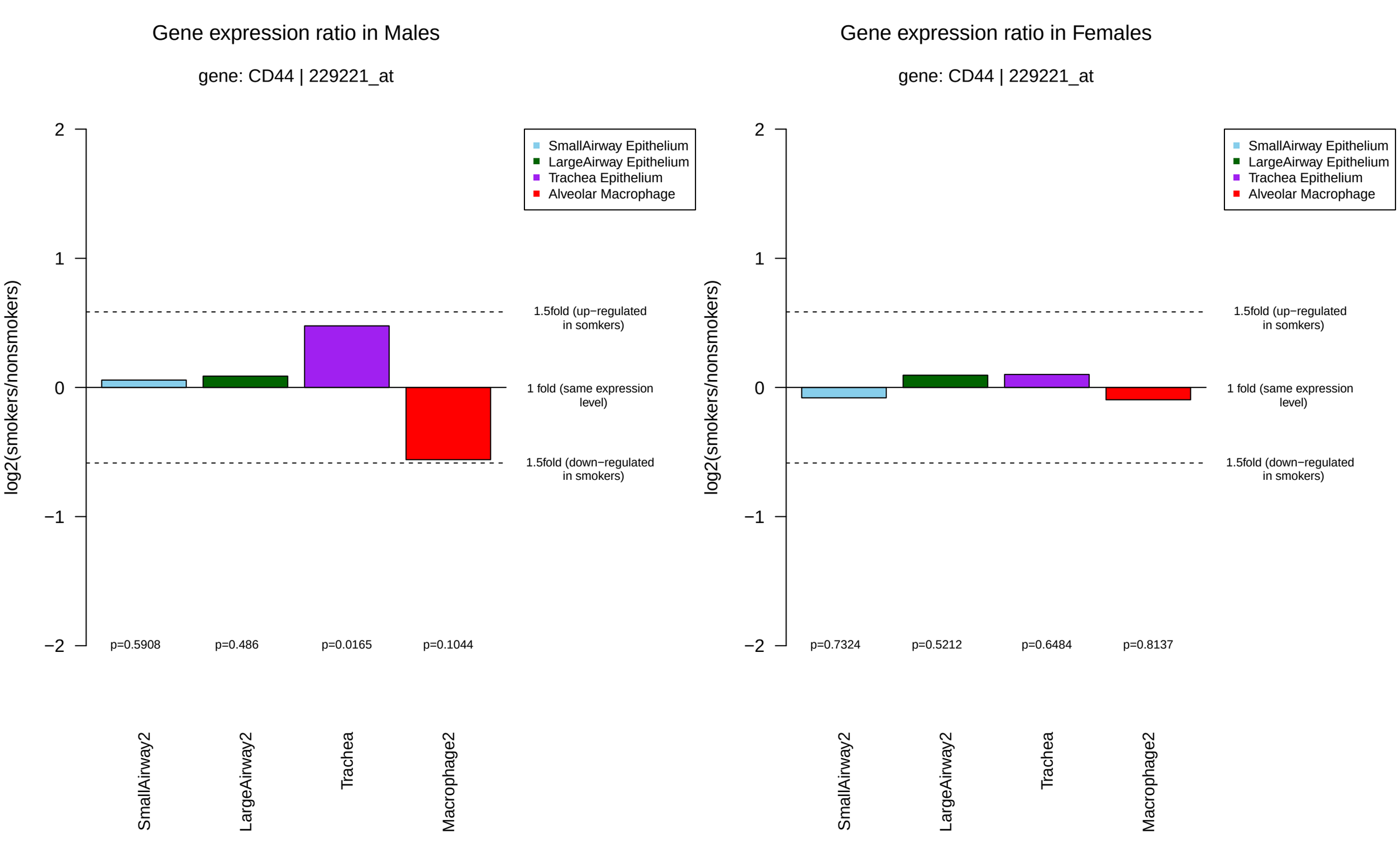
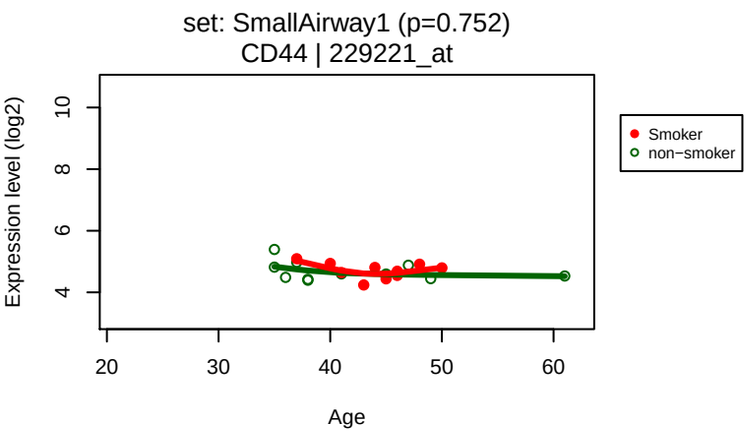
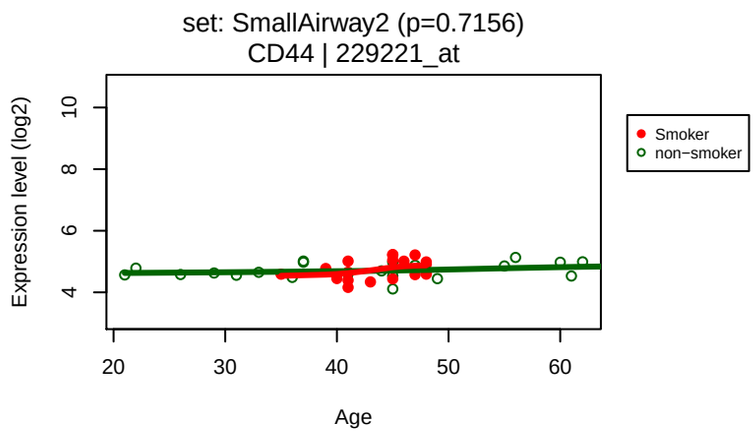
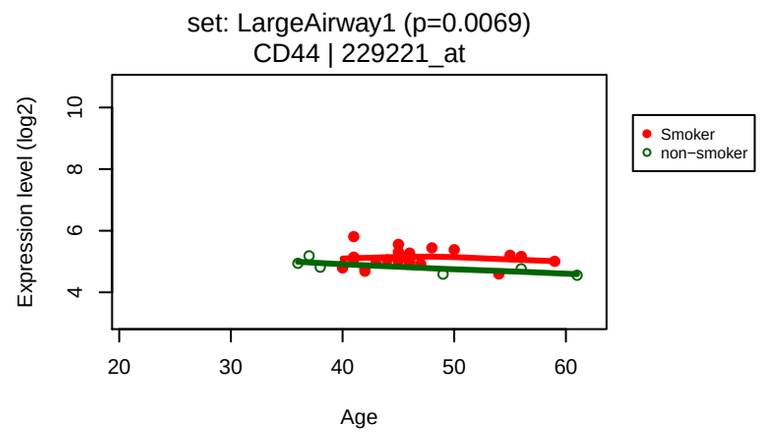

Gene expression of smokers and nonsmokers
gene: CD44 | 229221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.752
p=0.7156
p=0.1697
p=0.0069
p=0.2926
p=0.013
p=0.0121
p=0.1116
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

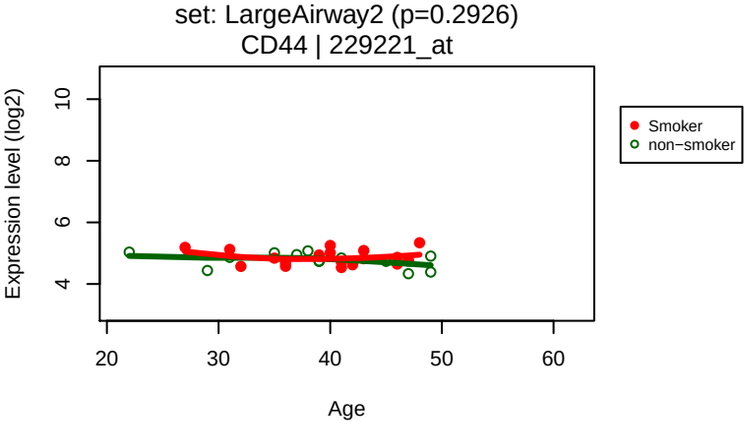

Gene expression ratio (smokers/nonsmokers)
gene: CD44 | 229221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.752
p=0.7156
p=0.1697
p=0.0069
p=0.2926
p=0.013
p=0.0121
p=0.1116
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

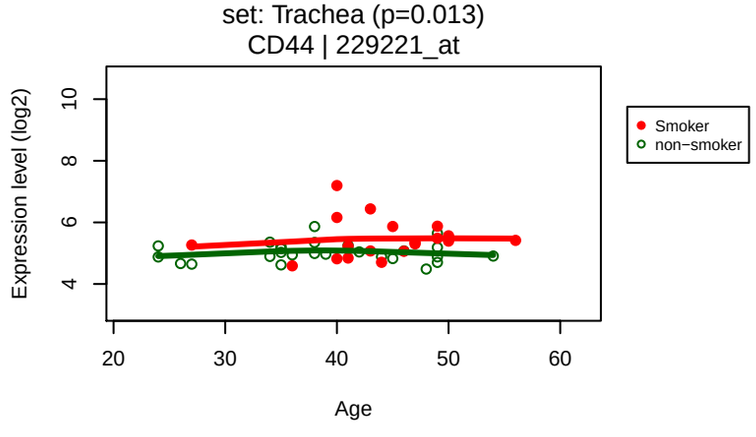

Gene expression ratio in Males
gene: CD44 | 229221_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5908
p=0.486
p=0.0165
p=0.1044
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CD44 | 229221_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7324
p=0.5212
p=0.6484
p=0.8137
SmallAirway2
LargeAirway2
Trachea
Macrophage2

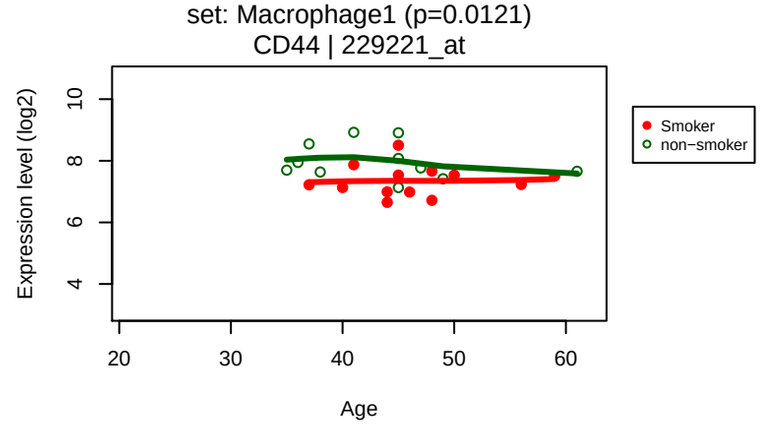

set: SmallAirway1 (p=0.752)
CD44 | 229221_at
Expression level (log2)
Age
Smoker
non−smoker

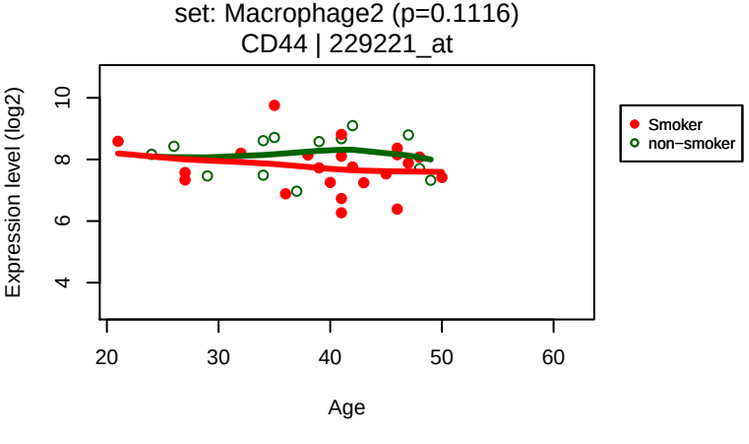

set: SmallAirway2 (p=0.7156)
CD44 | 229221_at
Expression level (log2)
Age
Smoker
non−smoker

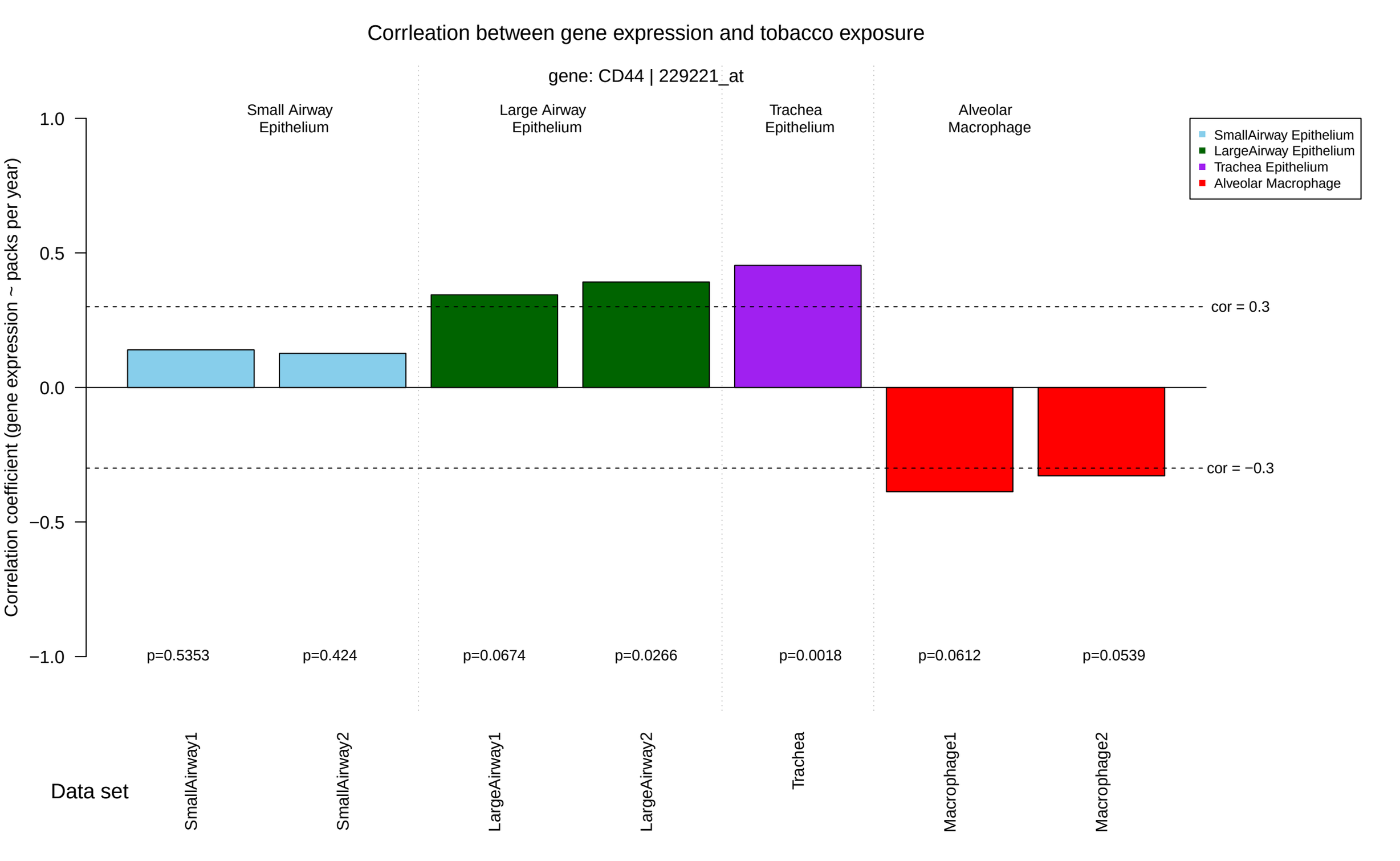

set: LargeAirway1 (p=0.0069)
CD44 | 229221_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2926)
CD44 | 229221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.013)
CD44 | 229221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0121)
CD44 | 229221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1116)
CD44 | 229221_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CD44 | 229221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5353
p=0.424
p=0.0674
p=0.0266
p=0.0018
p=0.0612
p=0.0539
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2